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ECON 5030

ECON5030 - Problem Set #2 - Game Theory

Q1)

	n_P	l_P	h_P
n_G	0,0	1,2	1,2
l_G	3,1	3,3	3,4
h_G	2,1	4,3	5,5

1A) round 1: greg: phil:

n_g strictly dominated by n_P s.d. by $l_G(3>0, 3>1, 3>1)$ $l_P(2>0, 3>1, 3>1)$

round 2:

	l_P	h_P
~~l_G~~	3,3	3,4
h_G	4,3	5,5

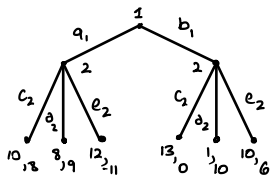
solution set is $\{h_g\} \times \{h_P\}$, payoffs $\{(5,5)\}$

1B) BR_G: $n_P \to l_g$, $l_P \to h_g$, $h_P \to h_g$

BR_P: $n_g \to \{l_P, h_P\}$, $l_g \to h_P$, $h_g \to h_P$

nash eq: $\{(h_g, h_P)\}$

2) Normal Form Games



2.2) $S_1 = \{a_1, b_1\}$

$S_2 = \{(0,0), (0,1), (1,0), (1,1)\}$

2.3)

	cc	cd	ce	dc	dd	de	ec	ed	ee
a_1	10,8	10,8	10,8	8,9	8,9	8,9	12,-11	12,-11	12,-11
b_1	13,0	1,10	10,6	13,0	1,10	10,6	13,0	1,10	10,6

2.4) vs a_1, P_2 needs d_2 after a_1, then a_1 if $y = d_2$

vs b_1, P_2 needs d_2 after b_1, then $b_1 \to (1 < 10, 8, 12)$

nash equilibrium: $(a_1, d_2 d_2)$

3) Timing & Strategies

3.1) no timing → (strategy = action)

3.2) $S_1 = \{0,1\}$

$S_2 = \{(0,0), (0,1), (1,0), (1,1)\}$

3.3)

	(0,0)	(0,1)	(1,0)	(1,1)
0	1,0	1,0	0,1	0,1
1	0,1	1,0	0,1	1,0

4) Third Party Observations

4.4) No

payoff = own $ → (M,M) not NE

vs M_1 deviate to L_1 which rejects

max own $ + NE

4.5) game def.

$G = (N, \{S_i\}, \{u_i\})$

contains:

- players
- strategies S_i (complete contingent plans)
- preferences u_i

  $u_i : \Pi_j S_j \to \mathbb{R}$: i's prefs over induced outcomes

4.6) fairness / social preferences:

$u_i = x_1 + x_2 + \min\{x_1, x_2\}$

	L_1	M_1	H_1
L_2	24,24	50,50	17,17
M_2	50,50	54,54	28,28
H_2	17,17	28,28	15,15

M strictly dominant

2.7) rationality test:

no, utilities are unobserved, so the data only tests rationality jointly w/ an assumed utility

5) Subgame Perfect Nash Equilibrium

5.1) $S_1 = \{L_1, M_1, H_1\}$

$S_2 = \{U, D\}^3$

$x_L x_M x_H$: action after L_1, M_1, H_1

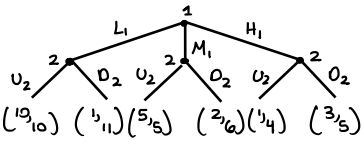
$u_i(K_1, x_L x_M x_H) = u_i(K_1, x_K)$

5.2)

	UUU	UUD	UDU	UDD	DUU	DUD	DDU	DDD
L_1	10,10	10,10	10,10	10,10	1,11	1,11	1,11	1,11
M_1	5,5	5,5	2,6	2,6	5,5	5,5	**2,6**	**2,6**
H_1	1,4	3,5	1,4	3,5	1,4	3,5	1,4	**3,5**

D are the 2 nash equilibriums

5.3)



P2: select D_2 @ every 2 node

P1: 1, 2, 3 → H_1

Subgame perfect nash equilibrium:

(H_1, DDD), payoffs $(3,5)$